Exercise 1: Optical Flow

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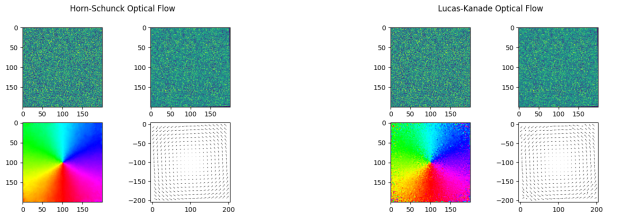
I. Introduction

In this assignment we compare two methods for estimating optical flow: Lucas-Kanade and Horn-Schunck. We test them on different types of images and analyze their performance.

II. Experiments

A. Testing on different types of images

First, we tested the methods on an image of random noise that has been rotated by 1 degree. The results are visible in figure 1. HS produces smooth flow due to the global regularization term, while LK appears smooth here, because random noise has gradients in all directions, which means it can calculate flow everywhere.



(a) Horn-Schunck method. (b) Lucas-Kanade method.

Figure 1: Optical flow estimation on rotated random noise.

We also tested the methods on other types of images. In figure 2 we can see that both methods work well when the small motion assumption holds. They both show motion concentrated around object boundaries and textured regions. HS is visibly smoother than LK, which is due to the global smoothness constraint. In the next example, seen in figure 3 LK estimates motion correctly only in the corners, where gradients exist in both spatial directions (for the rest they are 0). Along the edges motion is partially recovered, but it is not reliable. HS produces a smooth flow field, but it's not accurate. It shows motion everywhere and in random directions in some cases. This is the result of the aperture problem, where motion cannot be uniquely determined along edges.

The last example (figure 4) shows a non uniform motion due to perspective effects (the camera moving closer to the toy car). Both methods estimate the overall flow correctly by showing motion away from the center. Again the difference is that HS produces a smoother flow field, while LK shows more details but also more noise.

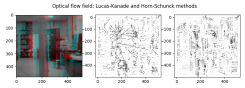


Figure 2: Optical flow estimation on translated image. Overlay of both images on the left, optical flow field of LK method in the middle and of HS method on the right.

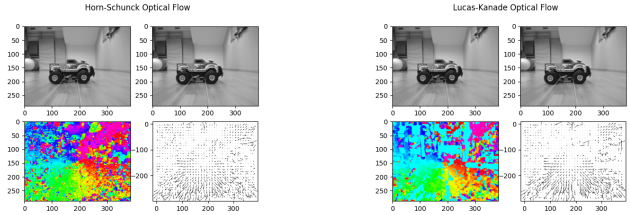
B. Reliability of Lucas-Kanade

In LK flow is computed only where the structure tensor has strong eigen values. To detect this we can use the Harris response detector. In figure 5 we plotted the magnitude of the flow computed by LK with and without Harris-based filtering. We can see that on the right the LK produces small noisy vectors in uniform areas, while on the left the harris filtering zeroed out pixels that the algorithm cannot reliably compute flow for (the motion has been preserved only in regions where the gradients are strong in both directions).



(a) Horn-Schunck method. (b) Lucas-Kanade method.

Figure 3: Optical flow estimation after shift to the right.



(a) Horn-Schunck method. (b) Lucas-Kanade method.

Figure 4: Optical flow estimation on collision images.

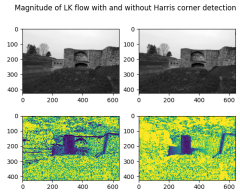
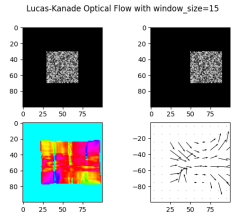


Figure 5: Comparison of LK with (left) and without (right) Harris detector.

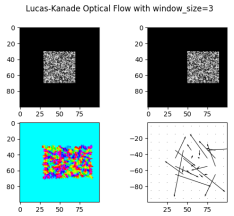
C. Parameters analysis

The LK method has a window size parameter which controls the trade-off between noise and detail. Larger windows may over smooth and lose fine details (as they average gradients over a larger region), while smaller windows can be more sensitive to noise. An example for this can be seen in figure 6. There is also the gaussian smoothing sigma parameter which controls how much the image is smoothed before computing the flow. A larger sigma can help reduce noise but can also blur important features. Lastly there is the harris threshold which if too high can discard good flow, while if too low can keep unreliable flow vectors.

The HS method has the smoothness regularization parameter λ. It's a weighing factor which decides how much importance do



(a) LK method with window size 15.



(b) LK method with window size 3.

Figure 6: Comparison of Lucas-Kanade with change in parameter.

we give to the brightness constancy constraint and how much to the smoothness constraint. A larger λ will produce smoother flow fields, but can also lose details (as it enforces stronger smoothness). On the other hand a smaller λ can capture more details but is more sensitive to noise. The difference is seen in figure 7. With $\lambda = 0.01$ the flow is very noisy because it prioritizes detail over smoothness, while with $\lambda = 10$ the flow is smoother but less accurate as it over-smooths the boundaries. The HS method also has the number of iterations parameter which controls how many times the flow is updated. More iterations leads to better convergence but also increases computational cost.

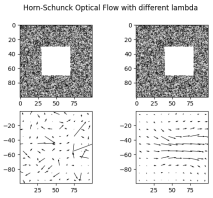


Figure 7: HS with $\lambda = 0.01$ on the left and $\lambda = 10$ on the right.

D. Runtime and initialization

The LK method is generally faster than HS since it computes flow only at certain points and uses a local window. The HS method uses iterative optimization which can be computationally expensive, especially for larger images. The runtime for both can be affected by the choice of parameters, such as window size for LK and number of iterations for HS. We tested the runtime for both methods on image pairs from the lab2 dataset. The results are shown in table I. We can see that LK is significantly faster than HS. However initializing HS with the output of LK can speed up convergence and reduce runtime.

The speed and performance of HS with and without harris detector is visualized in 8. The error is calculated as the sum of the data term and the smoothness term. Using LK initialization leads to faster convergence as it starts closer to the optimal solution.

Method	Average runtime (seconds)	Notes
Lucas-Kanade	0.0131	Fast, but can struggle with large displacements
Horn-Schunck	3.4208	Slower but more robust to noise
Horn-Schunck with LK initialization	2.9123	Faster convergence than HS alone by about 15%

Table I: Runtime comparison of optical flow methods

E. Pyramidal Lucas-Kanade

The pyramidal approach reduces large displacements by operating at a lower resolutions where motion appears smaller

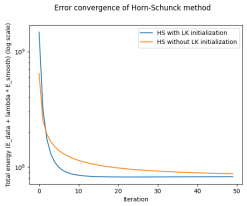
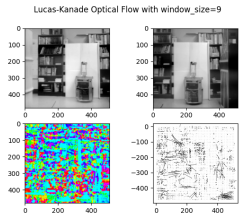
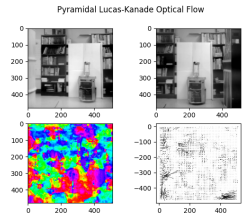


Figure 8: Error convergence of HS with and without LK initialization.

and then adjusting the estimate back to higher resolutions. The standard LK method works well only for small motions. If an object moves fast, the small motion assumption is violated. A pyramidal implementation addressed this issue by creating a pyramid of images at different scales. In figure 9 we can see how the standard LK fails to capture the motion of a fast moving object, while the pyramidal implementation is much more smooth and captures the motion much better.



(a) Lucas-Kanade method.



(b) Pyramidal LK method.

Figure 9: Comparison of LK and Pyramidal LK.

Running LK multiple times at the same scale is not a substitute for the pyramidal approach. To test this we ran 5 iterations of LK at the same scale but used warped images based on the previous flow estimate. We used two images from the provided collision image set that were 10 frames apart. In figure 10 we compare angle magnitudes of the flow estimation for the basic LK, iterative LK and pyramidal LK. We can see that the iterative approach slightly improves on the basic LK, but still fails to capture larger displacements (inconsisten colors in the plot, some white backgroung meaning zero motion or noise). So it confirms that the pyramid structure is the only robust way, as it has uniform colors around the expanding objects showing that it captures the consistent direction away from the center well.

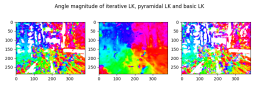


Figure 10: Angle magnitude of iterative, pyramidal and basic LK.

III. Conclusion

Lucas-Kanade is fast and captures local motion accurately, but it can struggle with large displacements. Horn-Schunck is can reduce noise and capture global motion, but it is computationally expensive and can blur motion boundaries. Adding Harris-based filtering improves LK reliability, while initializing HS with LK output can speed up convergence. The pyramidal approach is essential for handling large motions effectively. Overall, the choice of method depends on the task at hand.